

30. The smoke emanating from ammonium chloride powder used for cleaning hot utensils while galvanizing brass utensils is -

- (a) Ammonia
- (b) Carbon monoxide
- (c) Chloric acid of hydrogen
- (d) Ammonia and hydrochloric acid

46th B.P.S.C. (Pre) 2004

Ans. (d)

The smoke emanating from the ammonium chloride powder used for cleaning hot utensils when galvanizing brass utensils is ammonia and hydrochloric acid.

31. Radium was obtained from which mineral?

- (a) Limestone
- (b) Pitchblende
- (c) Rutile
- (d) Hematite

39th B.P.S.C. (Pre) 1994

Ans. (b)

Radium is obtained from pitchblende. Pitchblende is an ore of radium. But the proportion of radium in pitchblende is extremely low. Curie couple refined pitchblende to obtain radium.

32. Consider the following statements :

1. Crystallization is the process of deriving crystals of pure substance from their solution.
2. A bonding between constituent atoms or molecules takes place during crystallization.
3. It is an example of chemical change.

Which of the statements given above is/are correct?

- (a) 1, 2 and 3
- (b) 1 and 2 only
- (c) 1 only
- (d) None of the above

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Ans. (b)

Crystallization is a physical change that involves the formation of solids from a solution when atoms or molecules are organized into a crystal structure. Chemical bonds form between atoms or molecules during crystallization. Hence option (b) is the correct answer.

33. Which of the following substances will show Tyndall effect?

- (a) Dilute solution of NaCl
- (b) Milk
- (c) Starch solution
- (d) More than one of the above

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Ans. (d)

The Tyndall effect is the scattering of light by particles in a colloid, or a very fine suspension. It's also known as Tyndall scattering. Both milk and starch solutions are colloids. Thus, they will show the Tyndall effect.

III. BIOLOGY

Subdivisions of Biology

1. Pomology is the study of-

- (a) leaves
- (b) fruits
- (c) flowers
- (d) None of the above

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Ans. (b)

Pomology is a branch of botany that focuses on the science and practice of fruit production. The term comes from the Latin words pomum, meaning "fruit", and logy, meaning "study".

2. Histology is the study of

- (a) tissue
- (b) embryo
- (c) brain
- (d) neuron

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

Histology is the study of Tissue. Tissue is a group of cells that have similar structure and that function together as a unit. There are four basic types of Tissue-

1. Connective Tissue
2. Epithelial Tissue
3. Muscle Tissue
4. Nervous Tissue

3. What is Vermiculture?

- (a) The science of raising worms
- (b) The science of studying animals
- (c) The science of studying fishes
- (d) The science of killing worms

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Vermiculture is the process of decomposing and cultivating earthworms. Earthworms can help turn organic waste into nutrient-rich soil of garden.

4. Genetics deals with-

- (a) Mendel's laws
- (b) Organic evolution
- (c) DNA structure
- (d) Heredity and variations

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Genetics is the study of genes, heredity and genetic variations in living organisms.